

Heterogeneous Treatment Effect Models and Inference

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Causal Inference Series

1. Foundations
2. Defining some Causal Models
3. Inference, Asymptotic Theory, and Bootstrapping
4. Best Practices: Outliers, Feature Selection, and Bad Control
5. Heterogeneous Treatment Effect Models
6. Arguable Validation
7. Panel Data
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Overview

- This presentation covers the general problem of estimating heterogeneous treatment effects (HTE) and how it differs from ATE/ATET estimation.
- Covers a few models:
 - Linear models via Double Machine Learning
 - Causal Forests / Local Linear Forests
 - Doubly Robust models following Kennedy (2020)
- Wrap up with a simulation demonstration
- Mentions about extensions to panel models

HTE Overview

- "What's the effect of faster shipping?" Average treatment effect (ATE) and average treatment effect on the treated (ATET) models want to know aggregate treatment effects.
- "What's the effect of faster shipping on customer in rural areas? / Which customers benefit the most from faster shipping?" HTE model want to estimate the distribution of treatment effects or for a specific subset or individuals

$$Y_i = \hat{\beta}X_i + \hat{\tau}(Z_i)W_i + \epsilon_i$$

- $\hat{\tau}(Z_i)$ is the HTE and it varies of Z_i . We keep X_i different from Z_i for more flexible notation.
- We can also call this the conditional average treatment effect

HTE as an estimated function

- We want to estimate the functional form of HTE.
- When estimating ATE/ATET, we are only concerned with the average.
 - We average over more granular treatment effects.
- Estimating more granular treatment effects means there are additional challenges.

How much variation do we want in our HTE function?

- Two extremes:
 1. Individualized treatment estimates allow more flexibility, but can demand large sample sizes and variation in data.
- Increases the risk of noise driving estimates
- 2. Segmented estimates are the least flexible, with the least risk of noise driving estimates.
- They can also be more interpretable for other applications. • In-between case is to allow treatment effects to vary across some dimensions, but not others

HTE ideal experiment

- We can understand these two extremes based on what the ideal experiment is to estimate unbiased HTE.
- For individualized HTE, the ideal is to randomize treatment for each individual. (impossible)
- For segmented HTE, the ideal is to randomize treatment for each segment. (stratified randomization)
- The more individualized HTE is, the more data and assumptions are needed to distinguish between real patterns and statistical noise in the data.

HTE inference challenge

- Statistical inference for ATE/ATET estimates is based on the distribution of error around the average estimate.
- The challenge is getting a distribution around an individual estimate.
- The solution is to rely on either model specifications or bootstrapping-esque methods.

Some HTE Models

Support across use cases

| | Cross Sectional Data | Panel Data | Continuous Treatment

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Generalized Random Forests	Y	N	N

|Doubly Robust | Y | N | N

DML-Style models

- Semenova, Goldman, Chernozhukov, Taddy (2021) - SGCT
- Let's start with linearity assumptions, which allows interpretability:

$$Y_i = \hat{\beta}X_i + \hat{\tau}(Z_i)W_i + \epsilon_i$$

- SGCT decomposes $\hat{\tau}(Z_i)$ into a functional form:

$$\hat{\tau}(Z_i) \rightarrow \hat{\tau} \cdot g(Z_i)$$

- where $g(Z_i)$ is different functions of Z_i . For example:

$$\hat{\tau} \cdot g(Z_i) = \hat{\tau}_0 + \hat{\tau}_1 z_{1i} + \hat{\tau}_2 z_{1i}^2$$

- Continuing this example, the model would be:

$$Y_i = \hat{\beta}X_i + \hat{\tau}_0 W_i + \hat{\tau}_1 z_{1i} W_i + \hat{\tau}_2 z_{1i}^2 W_i + \epsilon_i$$

SGCT users residualization

- Now how do we estimate this equation?

$$Y_i = \hat{\beta}X_i + \hat{\tau}_0 + \hat{\tau}_1 z_{1i} + \hat{\tau}_2 z_{1i}^2 + \epsilon_i$$

- At first glance we can just do OLS, but we can improve that approach with double machine learning (DML; aka residualization).
 - Recall DML works through the Frisch-Waugh-Lovell theorem
- SGCT estimates this equation

$$\tilde{Y}_i = \hat{\tau}_0 \tilde{W}_i + \hat{\tau}_1 z_{1i} \tilde{W}_i + \hat{\tau}_2 z_{1i}^2 \tilde{W}_i + \epsilon_i$$

- Where $\tilde{Y}_i$ and $\tilde{W}_i$ are the residualized outcome and treatment

SGCT – HTE and inference

- We now need to do inference for individual treatment effects from

$$\tilde{Y}_i = \hat{\tau}_0 \tilde{W}_i + \hat{\tau}_1 z_{1i} \tilde{W}_i + \hat{\tau}_2 z_{1i}^2 \tilde{W}_i + \epsilon_i$$

- HTE is $\hat{\tau}_1 z_{1i} + \hat{\tau}_2 z_{1i}^2$, where the standard error is calculated via the Delta method.
- We can use OLS to estimate the above equation if:
 - • There are few dimensions of heterogeneity (ie $g(Z_i)$ is low dimensional); or
 - We are interested in specific dimensions of heterogeneity (ie we only want to know HTE across user tenure)

SGCT – inference with post-LASSO regression

- A problem is if we estimate with all possible transformations of z_{1i} . In other words, overfitting.
- We can select the relevant transformations of z_{1i} with LASSO regression, but then we cannot do inference.
- Get around this with a sample-split LASSO for inference. Select features with LASSO on one half of the dataset, and then estimate HTE using those selected features on the other half.

Generalized Random Forests

- Causal forests are a special class of generalized random forests, which we will discuss here.
- As motivation, note that under the unconfoundedness assumption, we assume that treatment is random:

$$E[(Y - \hat{g}(X_i) - \hat{\tau}(Z_i)W_i)W_i] = 0$$

- In other words, $\hat{\tau}(Z_i)$ satisfy the orthogonality assumption similar to Frisch-Waugh-Lovell.
- The problem is that we do not know what $\hat{\tau}(Z_i)$ looks like, so we want a flexible specification. Ideally, something non-parametric.

GRF – Causal Forest Objective Function

- The estimating equation (with simplified notation is):

$$(\hat{\tau}(Z_i), \hat{g}(X_i)) = \operatorname{argmin}\{E[\alpha_i(z)(Y_i - \hat{g}(X_i) - \hat{\tau}(Z_i)W_i|Z_i = z)^2]\}$$

- What's the purpose of $\alpha_i(z)$?
- $\alpha_i(z)$ is a weight used to allow flexibility in $\hat{\tau}(Z_i)$.
 - Can be estimated to kernel methods but performance suffers under high dimensions
 - Estimate $\alpha_i(z)$ with a random forest to deal with high dimensionality of Z_i !
- This weight gives us enough variation in $\hat{\tau}(Z_i)$.

GRF - Weights $\alpha_i(z)$

- $\alpha_i(z)$ represents the probability that a training sample i falls into the same leaf as sample z , across different trees in a random forest
- Splits in the random forest used to estimate $\alpha_i(z)$ are determined to maximize variation $\hat{\tau}(Z_i)$ across splits

GRF - Inference

- Athey, Tibshirani, and Wager (2019) show that $\hat{\tau}(Z_i)$ is asymptotically normal.
- This is because $\alpha_i(z)$ is estimated in an “honest” (Athey and Wager, 2018. fashion, where different samples are used to determine splits in $\alpha_i(z)$ and $\hat{\tau}(Z_i)$.
- Standard errors and confidence intervals are available based on a bootstrap/jackknife approach.
- Intuitively, estimate the distribution in $\hat{\tau}(Z_i)$ when z is removed from the sample

Doubly Robust - Kennedy (2020)

- Recall the interactive regression model from DML:

$$\hat{\tau}_{ATE} = E \left[(\hat{Y}_{1,i} - \hat{Y}_{0,i}) + \frac{W_i(Y_i - \hat{Y}_{1,i})}{\hat{P}_i} - \frac{(1 - W_i)(Y_i - \hat{Y}_{0,i})}{1 - \hat{P}_i} \right]$$

- Recall that we can intuitively understand this as a individual-level comparison from a regression adjustment model, correcting for prediction errors.
- Removing the expectation, we can see that these are individual-level treatment effect estimates

$$(\hat{Y}_{1,i} - \hat{Y}_{0,i}) + \frac{W_i(Y_i - \hat{Y}_{1,i})}{\hat{P}_i} - \frac{(1 - W_i)(Y_i - \hat{Y}_{0,i})}{1 - \hat{P}_i}$$

Applying inference to the individual estimates

- The problems are that these estimates:
 1. Are meant to be averaged to get the ATE/ATET; and
 2. Do not have inference properties.
- Kennedy (2020) frames these as “noisy” estimates of the true HTE, and proposes “refining” them with a second stage.

$$hte_i = (\hat{Y}_{1,i} - \hat{Y}_{0,i}) + \frac{W_i(Y_i - \hat{Y}_{1,i})}{\hat{P}_i} - \frac{(1 - W_i)(Y_i - \hat{Y}_{0,i})}{1 - \hat{P}_i}$$

Simulation Study

Context

- Recall, HTE models are not about estimating the average effect, but rather the functional form of treatment effects.
- The additional complexity of functional form can make this very difficult.
- We will demonstrate using simulation evidence, where we can change the **true HTE function**

General Simulation Context

- For simplicity, there is only one feature x

$$x \sim U[0, 1]$$

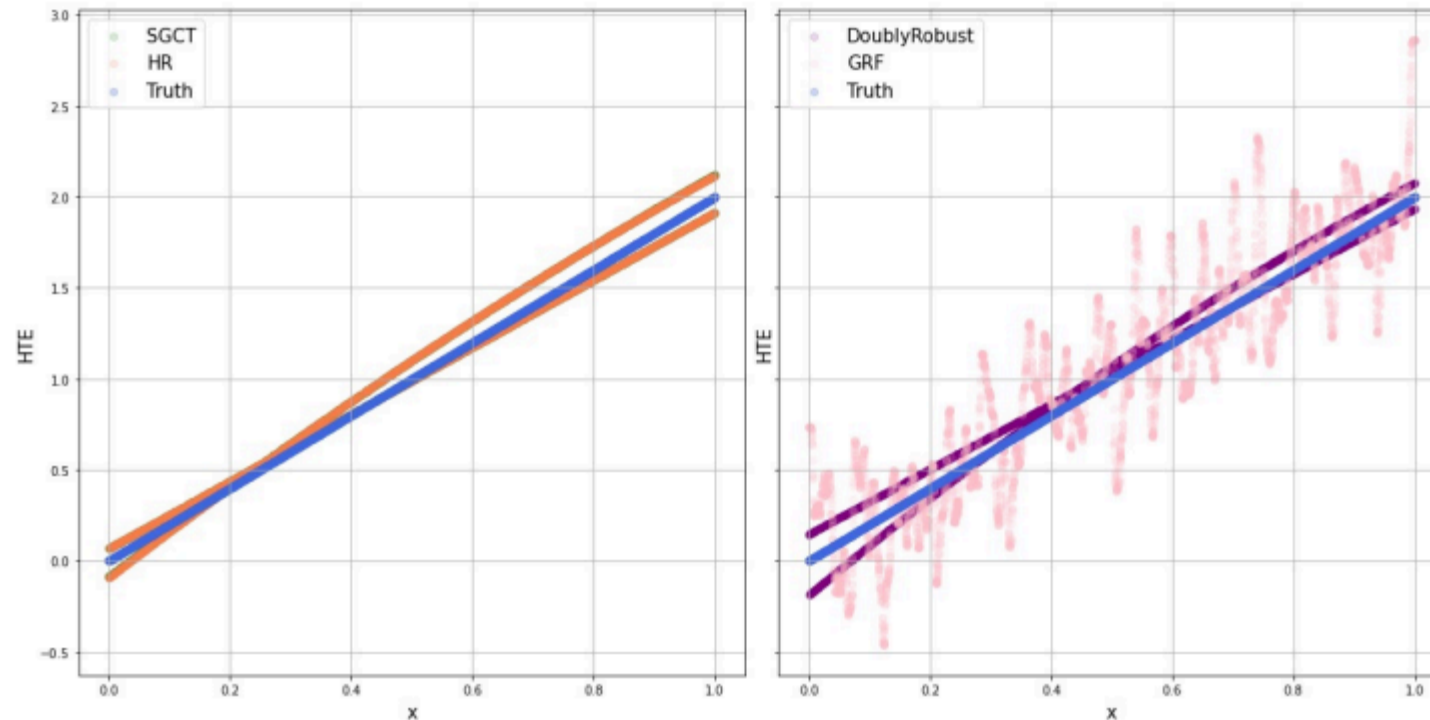
$$y = 10 + 2\ln(1 + x) + \epsilon, \epsilon \sim N(0, 1)$$

$$W = 1\left\{\frac{\exp(x)}{1 + \exp(x)} + \eta > 0\right\}, \eta \sim N(0, 1)$$

- We want to know the HTE of W_i
- We show three examples, with different HTE functions

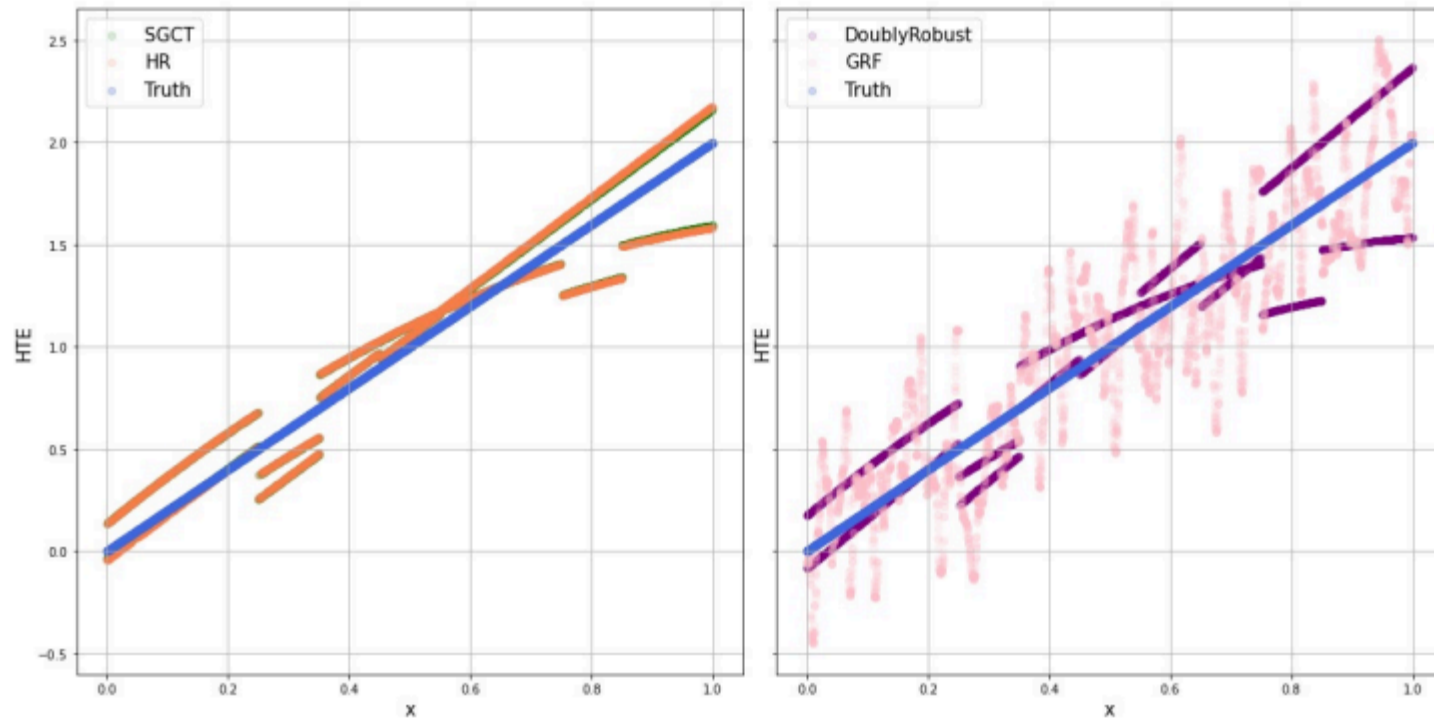
First Example - Linearity

- $HTE = 2x$
- Model controls: x, x^2



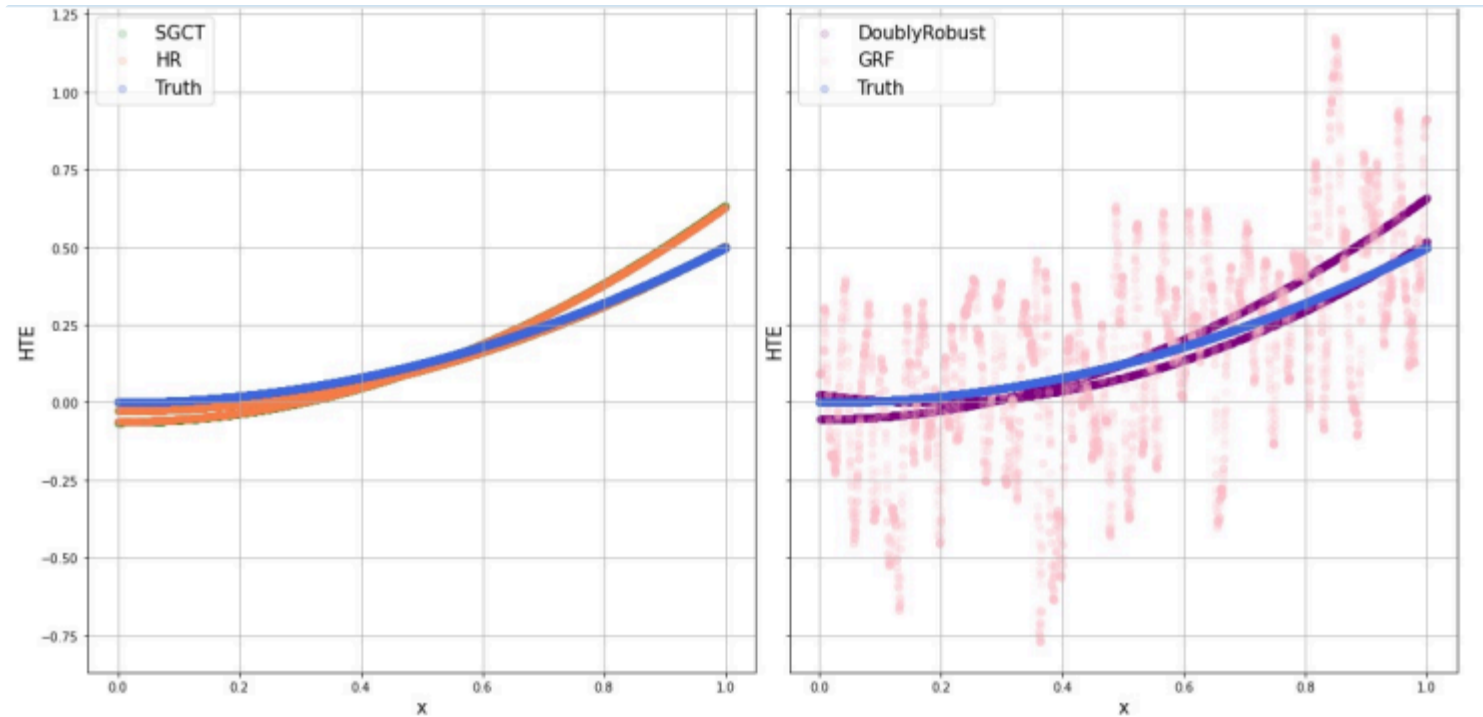
First Example - Linearity with more controls

- $HTE = 2x$
- Model controls: $x, x^2, 1\{0 \leq x < 0.2\}, \dots, 1\{0.8 \leq x < 1\}$



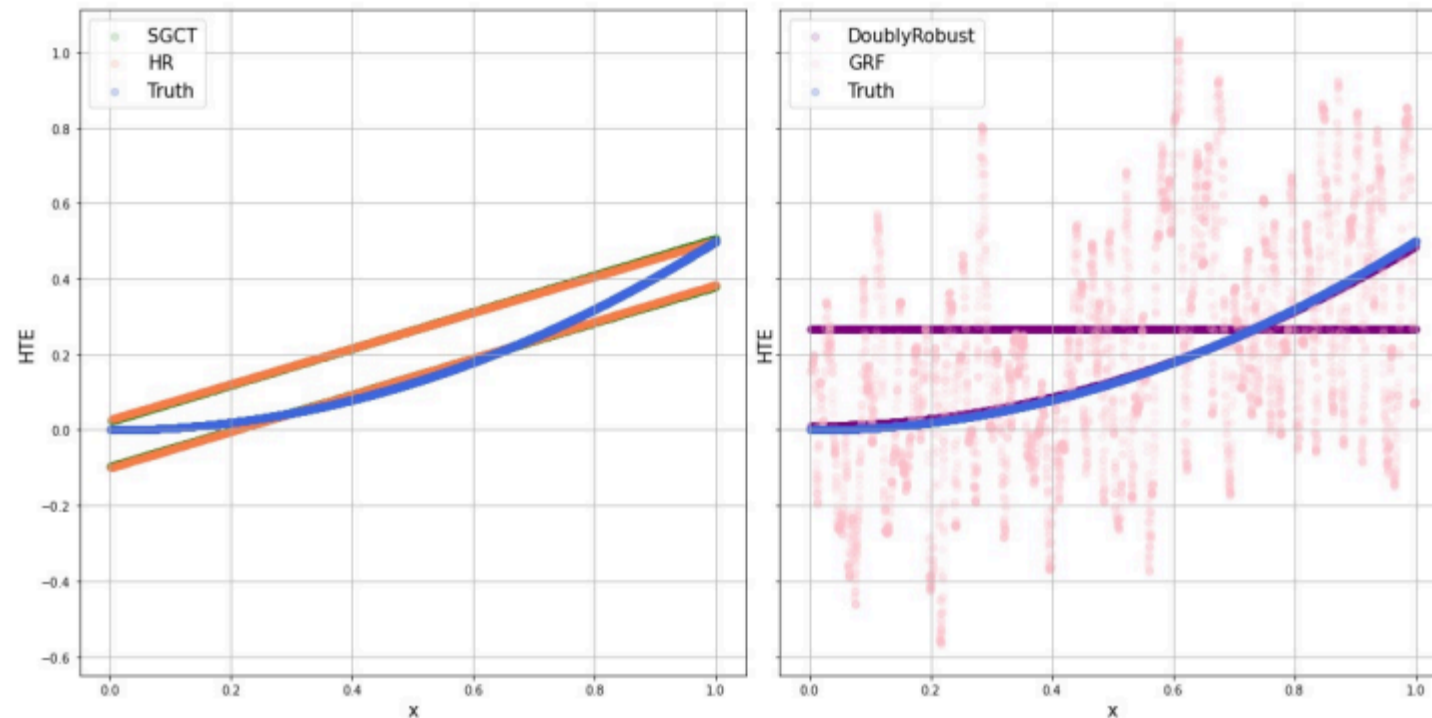
Second Example - Quadratics

- $HTE = 2x$
- Model controls: x, x^2



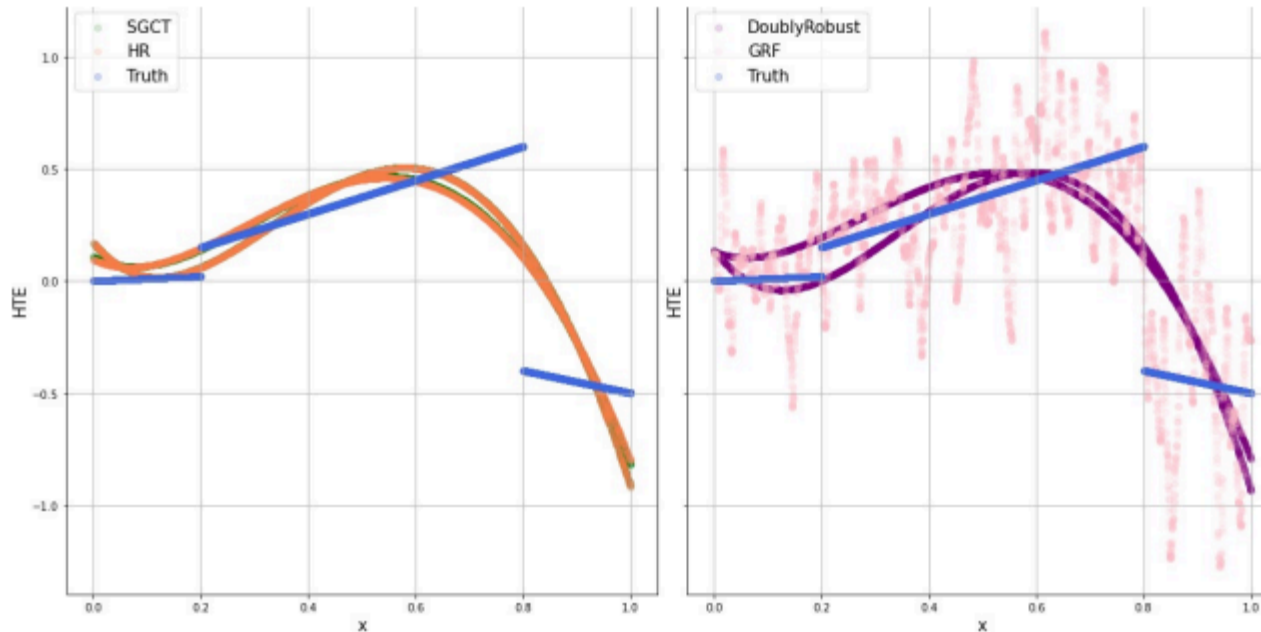
Second Example - Quadratics with more controls

- $HTE = 2x$
- Model controls: $x, x^2, 1\{0 \leq x < 0.2\}, \dots, 1\{0.8 \leq x < 1\}$



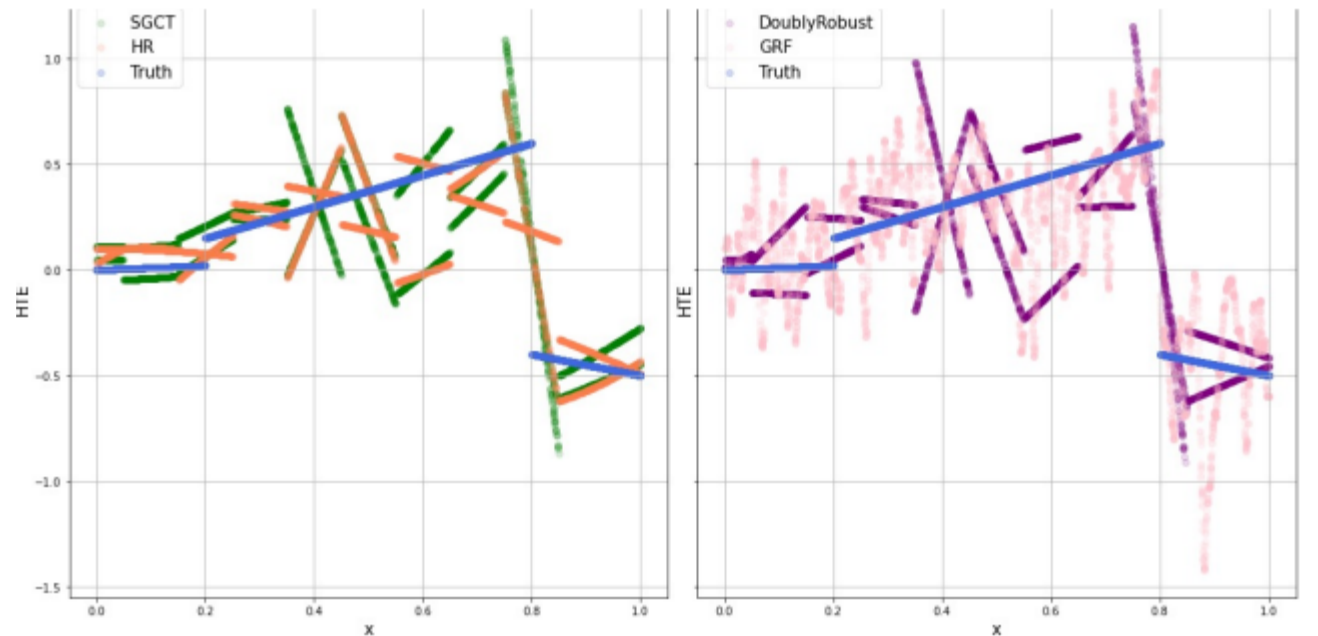
Third Example - Piece-wise

- $$HTE(x) = \begin{cases} 0.10x, & x < 0.20 \\ 0.75x, & 0.20 \leq x < 0.80 \\ -0.50x, & 0.80 \leq x \end{cases}$$
- Model controls: x, x^2



Third Example - Piece-wise with more controls

- $0.10x, \quad x < 0.20$
 $HTE(x) = 0.75x, \quad 0.20 \leq x < 0.80$
 $-0.50x, \quad 0.80 \leq x$
- Model controls: $x, x^2, 1\{0 \leq x < 0.2\}, \dots, 1\{0.8 \leq x < 1\}$



Takeaways

- Including more features to estimate a more flexible HTE may not necessarily increase performance.
- The more complicated, or more fine-grained, you want HTE estimates to be, the more data you need.

Review and Conclusion

- Covered the additional complexities and challenges of estimating HTE
- Covered a parametric (DML, HR) and non-parametric (forests) models
 - Deep neural network models (Farrell et. al 2020) not covered because of code availability
- Demonstration with simulated data

